TEK-UP

Time Series Course — Data Science & AI

MINI-PROJECT

Modeling, Forecasting and Diagnostics of Multidimensional Time Series

From classical ARIMA to neural networks, through cointegration, conditional volatility and generative AI

Two Themes to Choose From — Finance or Environment

This project can be carried out equally well on financial data (stocks, indices, currencies, cryptocurrencies) or on environmental and energy data (temperature, CO2, electricity, air quality). Both themes rely on exactly the same methodological toolkit — only the field of application differs. The grading scale and expectations are strictly identical for both choices.

Total grade	Submission deadline	Submission method
20 points	November 12, 2025, 11:55 PM	By email to ahmed.dhouibi@tek-up.de

Projects will be randomly selected for an oral defense to be held at a later date. Every student must therefore be able to explain and defend the entirety of their own work, including the code.

0. Introduction and Learning Objectives

This mini-project puts you in the position of a data scientist facing a real, non-trivial problem: choosing, fitting, diagnosing and comparing several families of time series models on data you will have collected and prepared yourself. This is not about applying a single recipe, but about building a reasoned modeling approach, where every choice (transformation, model order, network architecture) is justified by a statistical diagnostic or an empirical observation.

The project deliberately covers a wide spectrum — from classical econometric models (ARIMA, SARIMA, VAR, cointegration, GARCH) to recurrent neural networks and the reflective use of generative artificial intelligence. This diversity is not meant to overwhelm you: the grading scale rewards depth and rigor of diagnostics far more than the sheer number of stacked models. A well-diagnosed ARIMA, honestly compared to its baseline, is worth more than a poorly validated LSTM.

Targeted skills

- Build a complete time series processing pipeline, from raw data to evaluated forecast.
- Master the statistical diagnostics that justify choosing one model over another (stationarity, autocorrelation, cointegration, conditional heteroscedasticity).
- Rigorously compare econometric models and deep learning models on the same dataset, using the same metrics.
- Use an LLM as an analytical assistance tool, while maintaining a critical eye on its output.
- Write a scientific report that clearly articulates method, results and limitations.

1. Choice of Theme and Data Import

You are free to choose one of the two following themes. This choice is final and must be justified in one sentence in the report's introduction (why this series, why this theme).

1.1 Option A — Financial Data (Yahoo Finance)

The Yahoo Finance website (finance.yahoo.com) provides access to numerous multidimensional time series: stocks, stock market indices, currencies, cryptocurrencies. Each asset provides at minimum a multidimensional vector (Open, High, Low, Close, Adjusted Close, Volume).

- Choose one asset or a basket of 2 to 3 correlated assets (e.g. a stock and its sector index, or two cryptocurrencies), at daily frequency and covering at least 5 years.
- Data can be retrieved via the R package quantmod or tidyquant, or in Python via yfinance.
- Construct at least one derived technical indicator (moving average, realized volatility, log return) in addition to the raw variables.

1.2 Option B — Environmental and Energy Data

A pedagogically equivalent alternative to finance, often more intuitive to interpret and carrying different stakes (very pronounced seasonality, long-term trends, breaks linked to public policy). Recommended public sources:

Data type	Recommended source	Typical frequency
Temperature, precipitation	NOAA (ncdc.noaa.gov), ECMWF/Copernicus (Climate Data Store)	Daily to monthly

Data type	Recommended source	Typical frequency
Atmospheric CO2	NOAA Global Monitoring Lab (Mauna Loa), Our World in Data	Monthly
Electricity consumption and production	RTE éCO2mix (France), ENTSO-E Transparency Platform (Europe)	Hourly to daily
Air quality (PM2.5, NO2, ozone)	European Environment Agency (EEA), OpenAQ	Hourly to daily
Aggregated climate and energy indicators	Our World in Data (ourworldindata.org), Kaggle (climate datasets)	Variable

- Choose a multidimensional series: for example, hourly electricity consumption together with temperature and day of the week, or CO2 concentration together with an indicator of industrial activity.
- Recommended depth: at least 3 years of data to capture several complete seasonal cycles.

1.3 Preparation Common to Both Options

- Convert the data into a native time series object (ts, xts or tsibble in R; DatetimeIndex in Python with pandas), with an explicit time vector and correctly declared frequency.
- Split the sample chronologically (never randomly): 90% for training, 10% for testing.
- Precisely document the source, the period, and any transformation already applied by the data provider (e.g. dividend adjustments for finance, homogenization corrections for climate data).

2. Exploratory Analysis

- Visualize the series: trend, seasonality, visible breaks, scale comparison if several variables are involved.
- Produce and interpret ACF and PACF on the raw series and then on the differenced series — the interpretation must be written, not just the plot displayed.
- Perform a decomposition (classical additive/multiplicative, or STL) and comment on each extracted component.
- Detect outliers (rolling boxplot, IQR, or a dedicated statistical test) and propose a justified treatment (keep, correct, indicator variable).
- Formally test for stationarity (ADF, KPSS — both, since they have opposite null hypotheses and complement each other) and explicitly conclude on the required order of differencing.

3. Classical Modeling

This section is the econometric core of the project. Every model must be accompanied by its complete residual diagnostic — a model without diagnostics will not be graded, regardless of the apparent quality of its forecasts.

3.1 ARIMA and SARIMA

- Identify the orders (p,d,q) and, if seasonality is present, (P,D,Q,s) from the ACF/PACF and confirm using an automatic search (auto.arima in R, pmdarima in Python) — compare both approaches.

- Complete residual diagnostics: Ljung-Box test (independence), normality test (Shapiro-Wilk or Jarque-Bera), ARCH test (conditional homoscedasticity — see section 3.3).

3.2 Multivariate Models: VAR and Cointegration

If your series is multidimensional (which is expected under both options), you must move beyond univariate analysis to model the interactions between variables.

- Fit a VAR (Vector AutoRegression) model on at least two correlated variables from your dataset, selecting the order via an information criterion (AIC/BIC).
- Test for Granger causality between the VAR variables and interpret the result.
- Test for cointegration between the non-stationary series (Johansen test, or Engle-Granger test for two series): two series that are individually non-stationary may still share a long-run equilibrium relationship.
- If cointegration is detected, fit a Vector Error Correction Model (VECM) and interpret the error-correction term (speed of return to equilibrium).

Concrete Example by Theme

Finance: a stock and its sector index may be cointegrated — their long-run gap is stable even though each series individually follows a random walk.

Environment: electricity consumption and temperature may be cointegrated — their structural relationship (heating/cooling effect) persists beyond short-term fluctuations.

3.3 Conditional Volatility Models: ARCH and GARCH

Relevant as soon as the residuals of an ARIMA/VAR model fail the ARCH test (non-constant variance), which is very common in financial returns and not rare in extreme climate series.

- Check for ARCH effects on the residuals of the best ARIMA/VAR model obtained previously (ARCH-LM test).
- Fit a GARCH(1,1) model on the return series (finance) or on deviations from trend (environment) and compare with higher orders if the diagnostic justifies it.
- Diagnose the standardized residuals of the GARCH model (residuals should become homoscedastic again once volatility has been correctly modeled).
- Plot the estimated conditional volatility over time and relate it to known events in your series (financial crisis, heat wave, consumption peak).

4. Advanced Modeling

4.1 Recurrent Neural Networks

- Implement and compare at least two architectures: LSTM and GRU, with data formatting adapted accordingly (sliding windows, normalization).
- Justify hyperparameter choices (window size, number of layers, dropout) and document the training procedure (train/validation loss curves).
- Evaluate robustness through temporal cross-validation (time series cross-validation / walk-forward validation — never classical k-fold on a time series).

4.2 Hybrid Approaches

Optional but strongly rewarded: combine an econometric model and a deep learning model, for example by modeling trend/seasonality with ARIMA or Prophet and then modeling the remaining nonlinear residuals with an LSTM.

4.3 Additional Models of Choice (Deepening)

For students wishing to explore beyond the required core, the following options are rewarded as a deepening component (see grading scale):

Model	Pedagogical interest	Difficulty level
Prophet (Meta)	Automatic trend/seasonality/holiday decomposition, robust to missing data	Easy to implement
N-BEATS	Purely neural, interpretable architecture, without manual feature engineering	Intermediate
Temporal Fusion Transformer (TFT)	Multi-horizon attention, native handling of multiple exogenous variables	Advanced
Time series Transformers (Informer, PatchTST)	Recent state of the art for long series	Advanced — exploratory only

5. Generative AI and LLMs (4 points)

This section evaluates your ability to use an LLM (ChatGPT, Claude, Gemini or equivalent) as an analytical assistance tool, while maintaining a critical perspective — not as an automatic report generator that you would copy without understanding.

- Generate, from the descriptive statistics of your series (mean, variance, skewness, kurtosis, results of stationarity tests), model hypotheses to be tested by the LLM. Document the prompt used and the response obtained in full.
- Have the LLM explain your model results in a simplified way, as if the report were addressed to a non-statistician decision-maker. Include this explanation as an appendix.
- Generate simulated investment recommendations (finance option) or management/policy recommendations (environment option, e.g. anticipating a consumption peak) from your forecasts — systematically making explicit the limitations and risks of these recommendations (the LLM must be prompted to formulate them itself, and you then complement them with your own critical assessment).
- Compare the consistency between your own expert interpretation (written before consulting the LLM) and the explanation generated by the LLM: where do they converge? Where does the LLM get it wrong, oversimplify, or invent an unfounded causal relationship?

Beware of Statistical Hallucination

General-purpose LLMs can produce statistical claims that sound plausible but are false (invented p-values, unverified causal claims). Part of the grade for this section specifically assesses your ability to detect and flag these errors in the LLM's output, not merely to report them.

6. Forecasting and Evaluation

- Produce forecasts at horizon h = 5 to 10 steps for each retained model, with confidence/prediction intervals whenever the model allows it.
- Compare against the actual observations of the test sample using the MSE, RMSE, MAE and MAPE metrics — presented in a single summary table grouping all tested models.
- Test whether performance differences between models are statistically significant (Diebold-Mariano test) rather than merely comparing raw metric values.
- Explicitly conclude on the best model for your use case, taking into account not only performance but also interpretability and computational cost.

7. Possible Extensions (Bonus)

These extensions are not mandatory but are rewarded in the grading scale (see Evaluation section). They can be adapted to your theme:

- Introduce exogenous variables external to your main series: economic news or sentiment indices (finance), or complementary weather indicators and public holidays (environment).
- Apply event detection using text embeddings (transformer models such as BERT/FinBERT) on a corpus of news or reports related to your series, and link these events to the observed breaks.
- Study the dynamic correlation between two series over time (rolling correlation, DCC-GARCH) rather than a single static correlation — for example a stock and its sector index, or temperature and electricity consumption across seasons.

8. Expected Deliverables

8.1 Complete Code

- Code in BOTH R and Python for at least the classical core (section 3) — the advanced core (section 4) and the extensions may be implemented in only one of the two languages, at your choice, but must then be clearly identified as such.
- Commented code: each block must explain in one line what it does and why this choice was made at this step (not a paraphrase of the code, a justification).

8.2 Detailed Report (Word or PDF)

- Introduction to the problem: chosen theme, selected series, modeling question posed.
- Analysis steps, plots and diagnostics, presented in the logical order of the project (sections 1 to 6).
- Detailed model comparison table (metrics, complexity, interpretability).
- Interpretation of results and the specific contribution of LLMs (what they helped gain in time or clarity, what they failed to do).
- Indicative length: 15 to 25 pages excluding appendices and code.

8.3 Executable Notebook

RMarkdown (.Rmd) or Jupyter (.ipynb), allowing the entire study to be reproduced end-to-end without any manual intervention other than running the cells. The notebook must execute without errors on a clean machine that has the packages listed at the top installed.

8.4 Critical Discussion

- Limitations of the models used (violated assumptions, insufficient sample size, unstable estimates, etc.).
- Advantages and drawbacks of using generative AI for time series analysis in your field of application — beyond the specific comparison requested in section 5, a broader reflection is expected here.

9. Detailed Grading Scale (20 points)

The grading scale is split into three blocks. The "Deepening and Extensions" block rewards methodological depth (cointegration, well-diagnosed GARCH, hybrid approaches, extensions from section 7) rather than the mere accumulation of models.

Block	Content	Points
A — Fundamentals	Data preparation, exploratory analysis, ARIMA/SARIMA/VAR with complete diagnostics, forecasting and comparative evaluation (sections 1, 2, 3.1-3.2, 6)	10
B — Advanced Modeling	GARCH with diagnostics, LSTM/GRU with temporal cross-validation, rigorous classical vs. advanced comparison (sections 3.3, 4.1)	4
C — Generative AI and LLMs	Hypothesis generation, simplification, recommendations with explicit limitations, human/LLM comparison (section 5)	4
D — Deepening and Extensions	Cointegration/VECM, hybrid approaches, additional models (Prophet, TFT...), extensions from section 7	2

9.1 Cross-Cutting Quality Criteria (Applied to All Blocks)

Criterion	What is rewarded	What is penalized
Diagnostic rigor	Statistical tests correctly interpreted, assumptions verified before applying a model	Model applied without prior verification, test results not interpreted
Reproducibility	End-to-end executable notebook, fixed seed, documented package versions	Code that fails to run, dependency on undocumented local files or paths
Scientific honesty	Model limitations and failures explicitly reported, uncertainty quantified	Results presented as definitive without discussion of limitations
Critical thinking toward the LLM	LLM errors and approximations identified and corrected	LLM-generated content copied without critical review

Important Reminder

A sophisticated but poorly diagnosed model earns fewer points than a simple model that is perfectly diagnosed and honestly interpreted. Depth of reflection takes precedence over the number of algorithms used.

10. Practical Arrangements

Item	Detail
Submission deadline	October 12, 2026, 11:55 PM
Submission method	By email to ahmed.dhouibi@tek-up.de (code + report + notebook, .zip archive recommended)
Oral defense	Random selection of a subset of projects for a later oral defense — be prepared to defend the entirety of your own work
Individual or pair work	As specified in class — to be confirmed with the instructor if not otherwise stated

11. Suggested Resources

R Packages

forecast, fpp3 / fable, tsibble, urca (cointegration), vars (VAR), rugarch / fGarch (GARCH), keras / torch (LSTM/GRU), prophet, quantmod / tidyquant (finance).

Python Libraries

statsmodels (ARIMA, VAR, VECM), arch (GARCH), pmdarima, tensorflow/keras or pytorch (LSTM/GRU), prophet, yfinance (finance), pandas, scikit-learn (metrics, cross-validation).

Data Sources

Finance: finance.yahoo.com. Environment: NOAA (ncdc.noaa.gov), Our World in Data (ourworldindata.org), RTE éCO2mix, ENTSO-E Transparency Platform, European Environment Agency, OpenAQ.

Good luck to everyone.